

Symmetry and Conservation Laws in Modern Physics

X-yin

1

- $\Rightarrow$
- $\Rightarrow$
- $\Rightarrow$

2

- Lorentz Minkowski
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- Maxwell $U(1)$
- Yang-Mills $SU(2), SU(3)$
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3

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-
- $\Rightarrow$

4

$$Q = T_3 + \frac{Y}{2}$$

$Q = T_3 + Y$

5

$$\begin{aligned}\mathfrak{T} &= De \quad (\quad) \\ \mathfrak{R} &= D^2e \quad (\quad)\end{aligned}$$

6 $Q^2 = 0$

$$\begin{aligned}Q &= \partial, \quad Q^2 = 0 \Rightarrow \\ \star \int_{\partial\partial\Omega} (P \wedge \mathfrak{R}) &\equiv 0\end{aligned}$$

$$Q = d \quad \delta \Rightarrow \delta \mathbf{J} = 0$$

$$\mathbf{F} = d\mathbf{A}, \quad d\mathbf{F} = 0, \quad d \star \mathbf{F} = 4\pi \star \mathbf{J}, \quad \delta = \star d \star$$

$$\begin{array}{ccc} \mathbf{A} & & \\ \downarrow & & \\ \mathbf{F} = d\mathbf{A} & \xrightarrow{\star} & \star \mathbf{F} \\ \downarrow & & \downarrow \\ d\mathbf{F} = 0 & & d \star \mathbf{F} = 4\pi \star \mathbf{J} \xrightarrow{\delta = \star d \star} \delta \mathbf{F} = \mathbf{J} \\ & & \\ d^2 = 0 & & d \star \mathbf{J} = 0 \xrightarrow{\delta = \star d \star} \delta \mathbf{J} = 0 \end{array}$$

$$\delta^2 = 0$$

BRST

$$Q^2 = 0, \quad Q|\Psi\rangle = 0, \quad |\Psi\rangle \sim |\Psi\rangle + Q|\text{something}\rangle$$

7

- 1.
- 2.
- 3.

Einstein

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